

LOVELY PROFESSIONAL UNIVERSITY

Phagwara, Punjab, India

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

INTERNSHIP PROJECT REPORT

Al-Watania Information Systems, Riyadh, Saudi Arabia.

Project For Demonstration

MULTI-CLOUD ALERT SYSTEM

A Simulator for Cloud Infrastructure Monitoring and Security Alerting

Submitted by :

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Registration No : 12508688

B.Tech. Computer Science and Engineering

Minor: Cloud Computing

Internship Organization

WISYS, Riyadh, Saudi Arabia

Internship Role: IT Trainee

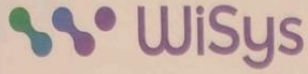
Internship Duration: 14 June 2026 – 13 August 2026

Company Supervisor

Emadeldin Taha Mahrous

Academic Session: 2026

INTERNSHIP COMPLETION CERTIFICATE



الشركة الوطنية لنظم المعلومات
AlWatania Information Systems

شهادة إتمام تدريب Training Completion Certificate					
Date: 13-Aug-2026			التاريخ: 29 صفر 1448		
Name :	Aqub Jawaid Ansari		عقوب جاويد انصاري		الاسم :
Passport:	AH565204		AH565204		جواز السفر:
Department :	Managed Services		إدارة الخدمات		الإدارة :
Training Period	Start Date	End Date	تاريخ البداية	تاريخ النهاية	فترة التدريب
	14-Jun-2026	13-Aug-2026	13-Aug-2026	14-Jun-2026	

AL-Watania Information Systems Company

الشركة الوطنية لنظم المعلومات

certifies that the mentioned Trainee has completed his Training with the company. During the period shown in the above schedule

تشهد بأن المتدرب الوارد ببياناته أعلاه قد أكمل تدريبه في الشركة خلال الفترة المذكورة في الجدول أعلاه

He proved to be hardworking, honest and punctual in his performance.
A high level of cooperation was exhibited in his Training at Wisys.

وكان المتدرب حسن السيرة والسلوك و مجتهداً في تأدية مهامه ومتعاون مع رؤسائه طيلة فترة تدريبه في الشركة.

This certificate was issued upon his request without any responsibility to the company to the purpose that may serve him best.

وقد أعطيت له هذه الشهادة بناء على طلبه دون أدنى مسؤولية على الشركة.

* (Any obliteration or alteration in the text of this certificate renders it cancelled).

* (أي شطب أو تعديل في نص هذه الشهادة يلغيها).

إدارة رأس المال البشري

Human Capital Management





Aquib Jawaid

IT Trainee

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AWS Cloud Practitioner Certification (during Internship)

aws  certified

Aquib Jawaid Ansari

AWS Certified Cloud Practitioner

VALIDATION NUMBER: 35fd0d001d0e48fe8718b038e0190ed5

VALIDATE AT: <https://aws.amazon.com/verification>

Issue Date: July 21, 2026

Expiration Date: July 21, 2029

DECLARATION

I, Aquib Jawaaid Ansari, student of B.Tech. Computer Science and Engineering, with a minor in Cloud Computing, hereby declare that this internship project report titled “Multi-Cloud Alert System” is prepared on the basis of the practical learning and work undertaken during my internship at WISYS, Riyadh, from 14 June 2026 to 13 August 2026.

The report is prepared for academic submission to Lovely Professional University, Phagwara. The Multi-Cloud Alert System presented in this report is a simulator developed to demonstrate the monitoring, alerting, audit-log and notification concepts related to the internship work.

No confidential credentials, customer information, private URLs, IP addresses, account identifiers, or other sensitive company information are disclosed in this report.

Student Signature: _____

Name: Aquib Jawaaid Ansari

Date: 20th August 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to WISYS, Riyadh, for providing me with the opportunity to complete my internship as an IT Trainee. The internship provided valuable practical exposure to cloud infrastructure monitoring, alerting, audit-log analysis and security-related activities.

I am especially thankful to my company supervisor, Emadeldin Taha Mahrous, for his guidance and support during the internship. I also express my gratitude to Lovely Professional University, Phagwara, and the School of Computer Science and Engineering for providing the academic framework and support required to complete the internship.

Finally, I am grateful to everyone who supported me during the internship and contributed to my learning and professional development.

CONFIDENTIALITY NOTICE

The internship involved work with organizational cloud environments and infrastructure. After completion of the internship, access to the company's cloud environments is no longer available. In addition, company security and confidentiality policies restrict the inclusion of production environment screenshots and sensitive infrastructure information in an academic report.

Therefore, this report does not include screenshots of the actual WISYS cloud environment, credentials, customer information, internal URLs, IP addresses, account identifiers or other confidential information. The practical demonstration in this report is provided through the Multi-Cloud Alert System simulator developed for academic purposes.

ABSTRACT

Cloud environments require continuous monitoring to detect infrastructure events, resource utilization issues and security-related activities. During my internship at WISYS, Riyadh, I gained practical exposure to cloud monitoring, infrastructure alerts, audit-log monitoring, IAM-related security events and cloud security assessment.

As an academic demonstration of this experience, I developed a Multi-Cloud Alert System simulator. The simulator represents a centralized monitoring approach in which cloud resources and events can be monitored and converted into alerts. It includes a dashboard for VM monitoring, alert generation, audit-log representation and real-time email notifications.

The project also documents the preparation of a GCP security report performed during the internship. The report presents the project at a non-confidential level and uses simulator screenshots rather than screenshots of the company's production environment.

Keywords: Multi-Cloud, Cloud Monitoring, Alerting, Security Monitoring, Audit Logs, IAM, VM Monitoring, Notifications

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1. INTRODUCTION

1.1 Background

Cloud computing enables organizations to deploy and operate infrastructure using scalable cloud resources. As cloud environments grow, continuous monitoring becomes necessary to maintain availability, identify operational problems and detect security-related changes.

During my internship at WISYS, Riyadh, I worked as an IT Trainee and gained practical exposure to cloud monitoring and security alerting activities. The internship work provided the foundation for the Multi-Cloud Alert System simulator described in this report.

1.2 Problem Statement

Monitoring resources across multiple cloud environments can become difficult when infrastructure events, resource thresholds and security events are handled separately. A centralized alerting approach can make important events easier to identify and communicate.

1.3 Project Objective

- To understand practical cloud infrastructure monitoring and alerting.
- To demonstrate monitoring of important VM and resource events.
- To represent security and audit-log based alerting.
- To provide centralized visibility of alerts from multiple cloud environments.
- To demonstrate real-time notification through email.
- To create an academic simulator that showcases the internship learning without exposing company data.

2. ORGANIZATION AND INTERNSHIP OVERVIEW

2.1 Organization

WISYS is an information technology organization based in Riyadh, Saudi Arabia. The internship provided practical exposure to cloud infrastructure operations, monitoring and security-related activities in a professional IT environment.

2.2 Internship Details

Student Name	Aquib Jawaid Ansari
Program	B.Tech. Computer Science and Engineering
Minor	Cloud Computing
Semester	5th Semester
School	School of Computer Science and Engineering
University	Lovely Professional University, Phagwara, India
Organization	WISYS
Location	Riyadh, Saudi Arabia
Role	IT Trainee
Internship Period	14 June 2026 – 13 August 2026
Company Supervisor	Emadeldin Taha Mahrous

2.3 Internship Work Area

The internship work included cloud infrastructure monitoring, alert configuration, event monitoring, audit-log analysis, security alerting and preparation of a GCP security report. The work involved understanding how operational and security events can be detected and communicated to responsible teams.

3. INTERNSHIP OBJECTIVES

- Gain practical experience in cloud infrastructure and IT operations.
- Understand how cloud resources are monitored.
- Understand VM lifecycle and resource utilization alerts.
- Learn how audit logs can be used for event and security monitoring.
- Understand IAM policy change monitoring.
- Understand the role of centralized monitoring across multiple cloud environments.
- Prepare and document cloud security findings and recommendations.
- Develop a simulator to demonstrate the cloud monitoring and alerting work performed during the internship.

4. PROJECT OVERVIEW – MULTI-CLOUD ALERT SYSTEM

4.1 Project Title

Multi-Cloud Alert System

4.2 Project Description

The Multi-Cloud Alert System is a simulator developed to showcase the cloud monitoring and security alerting concepts experienced during the internship. The simulator presents multiple cloud environments and provides a centralized view of infrastructure and security events.

The project is not the production monitoring system of WISYS. It is an independent academic demonstration that recreates the main concepts of the internship in a controlled environment. This allows the project to be demonstrated to the university without accessing or exposing the company's cloud environment.

4.3 Main Functions

- Cloud resource and environment representation.
- VM start monitoring.
- Infrastructure alert generation.
- Audit-log representation.
- Security-related alert display.
- Centralized alert dashboard.
- Real-time email notification.
- Alert and event information for investigation.

4.4 High-Level Workflow

Cloud Resources → Event Detection → Alert Generation → Central Dashboard → Email Notification

5. CLOUD MONITORING AND ALERTING WORK

5.1 Infrastructure Monitoring

Infrastructure monitoring was an important part of the internship. Monitoring allows operational teams to identify changes in resource state and utilization and respond to events in a timely manner.

5.2 VM Lifecycle Monitoring

VM lifecycle events are useful operational and security indicators. Events such as a VM being started, stopped, created or deleted can be monitored and converted into alerts.

5.3 Disk Utilization Monitoring

Disk utilization monitoring helps identify systems approaching a high storage utilization threshold. A configured threshold can be used to generate an alert when disk usage reaches the defined level.

5.4 Centralized Alerting

A centralized alerting approach provides a single place where infrastructure and security alerts can be reviewed. This reduces the need to manually check individual resources for every event.

6. SECURITY MONITORING AND GCP SECURITY REPORT

6.1 Audit-Log Monitoring

Audit logs provide information about actions performed on cloud resources. During the internship, audit-log information was used to understand and detect important operational and security events.

6.2 IAM Policy Change Monitoring

IAM policy changes are security-sensitive events because they can modify who has access to cloud resources. The monitoring approach focused on identifying policy changes and extracting useful event details such as the responsible identity, API method, service and affected resource where available.

6.3 GCP (Google Cloud Platform) Security Report

A GCP security report was also prepared during the internship. The report documented security findings and recommendations identified from the cloud environment. The findings were reviewed from a security and configuration perspective.

For confidentiality reasons, the detailed production findings, identifiers and environment screenshots are not reproduced in this academic report.

6.4 Security Monitoring Concept

Audit Event → Log Analysis → Security Condition → Alert → Notification

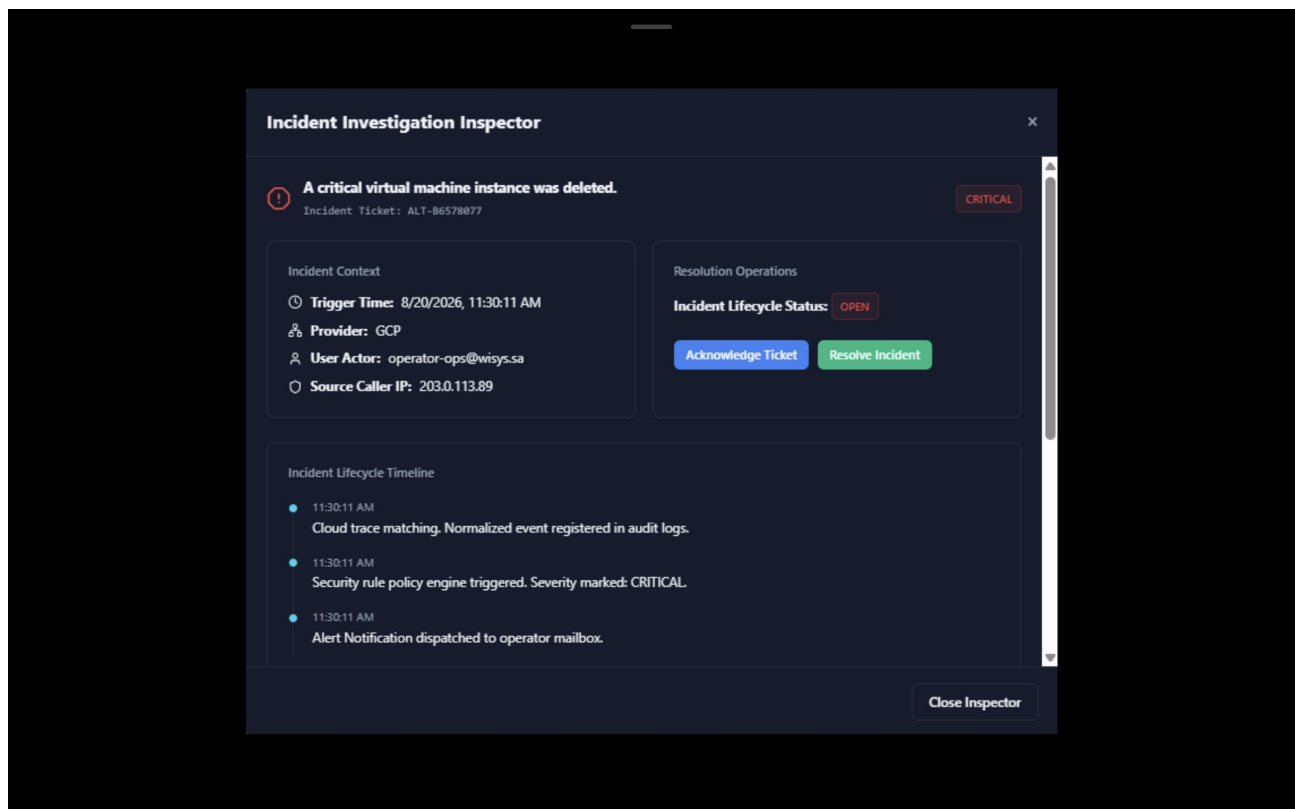
7. MULTI-CLOUD ALERT SYSTEM SIMULATOR

7.1 Purpose

The simulator was developed to provide a practical demonstration of the concepts learned during the internship. It allows the user to view monitoring information, generate alert scenarios, review audit events and receive real-time email notifications.

7.2 VM Deleted Dashboard

The VM deleted dashboard demonstrates the monitoring of virtual machine start events and provides a centralized view of the monitored environment.



7.3 Alerts

The alerts page displays generated events and provides a consolidated view of important infrastructure and security alerts.

WISYS MULTI-CLOUD
Operations & Alerting Console

Dashboard

Resources

Alerts 2

Event Simulator

Rules

Audit Logs

Security Score 85%

Notifications

CSE443 Internship Evaluation
Academic Local Sandbox

Console / Alerts

ACADEMIC SIMULATION ENVIRONMENT

Security Incident Alerts Console
Actionable alarms matched by the security rule engine

All Severities

All Providers

All Statuses

ALERT ID	CLOUD PROVIDER	CATEGORY	SEVERITY	RESOURCE	EVENT TYPE	DESCRIPTION	STATUS	ACTIONS
ALT-CB791087	GCP	COMPUTE	CRITICAL	vm-sap-prod	VM_DELETED	A critical virtual machine instance was deleted.	OPEN	Inspect <a>Acknowledge <a>Resolve
ALT-250EE746	GCP	COMPUTE	CRITICAL	vm-sap-prod	VM_STOPPED	A critical virtual machine instance was stopped.	OPEN	Inspect <a>Acknowledge <a>Resolve
ALT-78BAC345	GCP	COMPUTE	CRITICAL	vm-sap-prod	VM_STOPPED	A critical virtual machine instance was stopped.	RESOLVED	Inspect

7.4 Audit Logs

The audit-log page represents the type of event information used for investigating cloud activities. It provides visibility into the event details associated with monitored actions.

WISYS MULTI-CLOUD
Operations & Alerting Console

Dashboard

Resources

Alerts

Event Simulator

Rules

Audit Logs

Security Score 55%

Notifications

CSEA43 Internship Evaluation
Academic Local Sandbox

Console / Audit Logs

ACADEMIC SIMULATION ENVIRONMENT

System Audit Logs
Immutable ledger of administrative configuration changes and incident actions

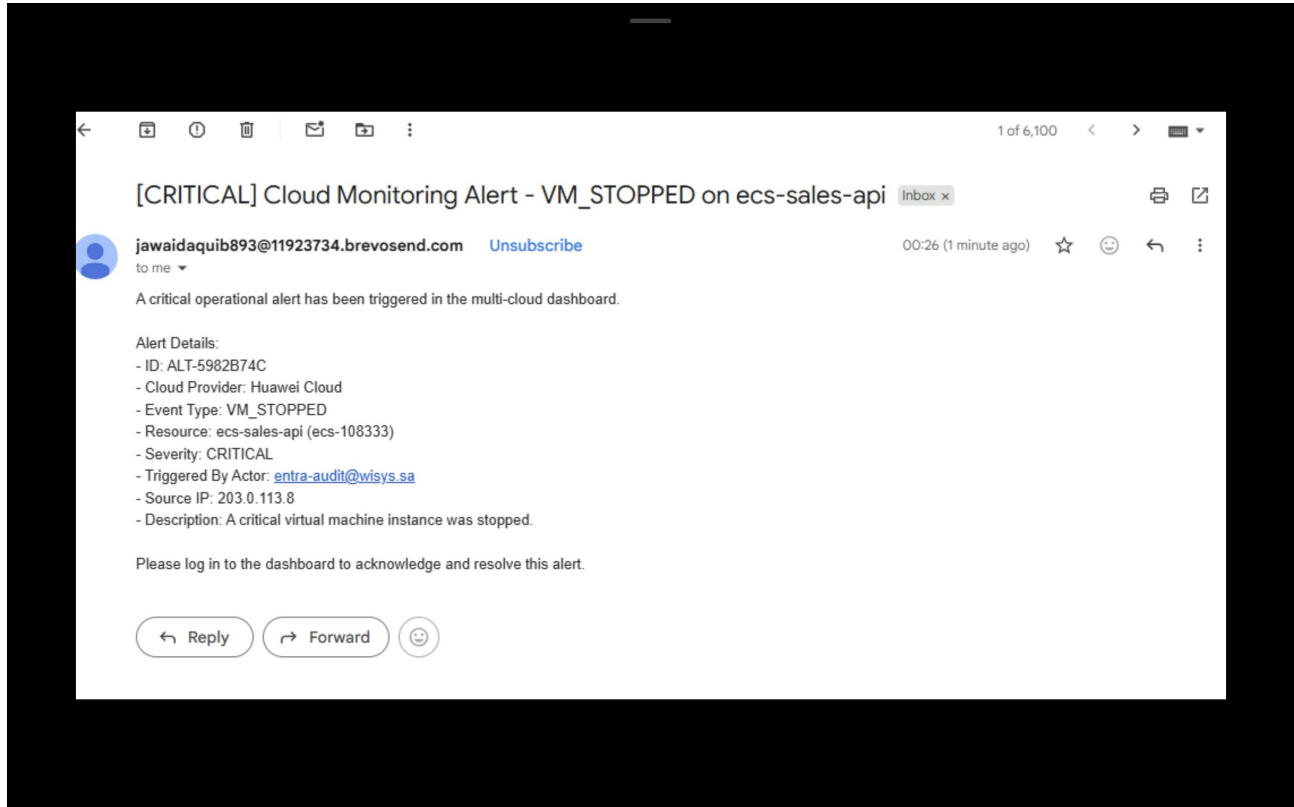
All Cloud Providers

Operations Access Log

TIMESTAMP	ACTOR / INITIATOR	CALLER IP ADDRESS	ACTION OPERATION	IMPACTED RESOURCE	CLOUD TENANT	STATUS CODE
8/19/2026, 6:56:36 PM	entra-audit@wisys.sa	192.0.2.25	Huawei CTS action deleteVpc executed by entra-audit@wisys.sa on vpc-hw-prod.	vpc-hw-prod	HUAWEI CLOUD	SUCCESS
8/19/2026, 6:56:23 PM	operator-ops@wisys.sa	283.0.113.89	Huawei CTS action deleteVpc executed by operator-ops@wisys.sa on vpc-hw-prod.	vpc-hw-prod	HUAWEI CLOUD	SUCCESS
8/19/2026, 6:56:16 PM	huawei-admin@wisys.sa	283.0.113.8	Microsoft Entra ID audit log: Update application - Certificates and secrets management on application WiSys Azure Sync Integration App.	WiSys Azure Sync Integration App	MICROSOFT ENTRA ID	SUCCESS
8/18/2026, 6:33:14 PM	entra-audit@wisys.sa	283.0.113.8	GCP event VPC_DELETED detected on resource vpc-gcp-production.	vpc-gcp-production	GCP	SUCCESS
8/18/2026, 6:32:03 PM	gcp-engineer@wisys.sa	192.0.2.25	GCP event VPC_DELETED detected on resource vpc-gcp-production.	vpc-gcp-production	GCP	SUCCESS

7.5 Notifications

The simulator can send real-time alerts through email. This demonstrates how a detected event can be communicated to an administrator or responsible team.

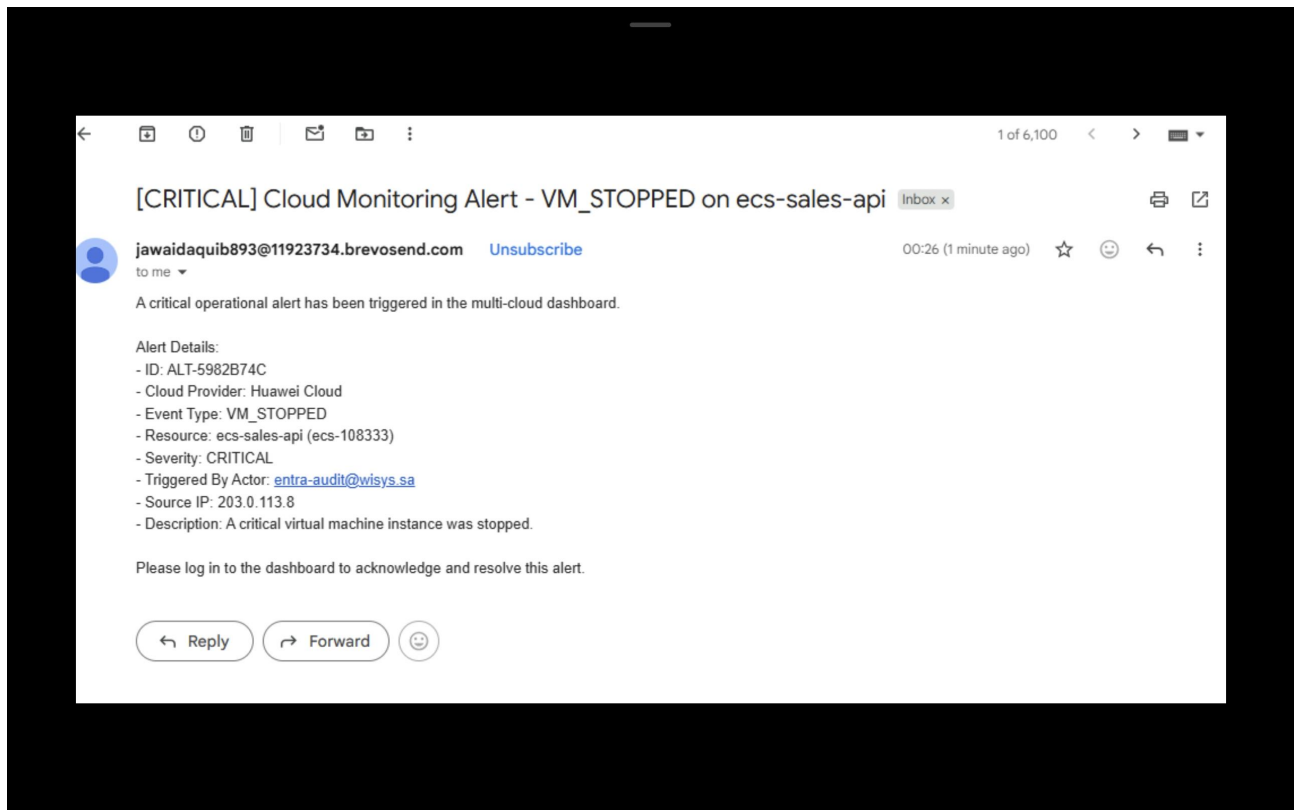


8. TESTING AND RESULTS

8.1 Test Cases

Test ID	Scenario	Expected Result	Result
TC-01	VM Stop	VM stop event is displayed	Pass
TC-02	Alert Generation	Alert is generated and displayed	Pass
TC-03	Audit Log	Relevant audit information is displayed	Pass
TC-04	Email Notification	Real-time email alert is received	Pass

8.2 Test Evidence



8.3 Results

The simulator demonstrates the intended monitoring workflow from event generation to centralized alert display and notification. The real-time email notification feature provides an additional demonstration of how alerts can be communicated outside the monitoring interface.

9. LEARNING OUTCOMES

- Practical understanding of cloud infrastructure monitoring.
- Understanding of VM lifecycle event monitoring.
- Understanding of resource utilization alerting.
- Understanding of audit logs and event investigation.
- Understanding of IAM policy change monitoring.
- Exposure to cloud security assessment and reporting.
- Understanding of centralized multi-cloud alerting.
- Improved technical documentation and troubleshooting skills.
- Experience in developing a simulator to demonstrate practical internship work.
- Awarded with AWS Cloud Practitioner Certification.

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

The internship at WISYS, Riyadh, provided practical exposure to cloud monitoring, alerting, audit-log analysis and security assessment. The work helped develop an understanding of how cloud infrastructure events and security-related activities can be monitored and communicated.

The Multi-Cloud Alert System simulator converts these practical concepts into an academic demonstration. It provides a centralized interface for monitoring scenarios, alerts and audit logs, along with real-time email notifications. Since the company environment cannot be shown due to confidentiality and security requirements, the simulator serves as the primary demonstration of the project.

10.2 Future Scope

- Integration with additional cloud providers.
- Integration with enterprise collaboration and notification platforms.
- Automated incident response.
- Advanced anomaly detection.
- Centralized security dashboards.
- Integration with SIEM platforms.
- Automated remediation of selected events.

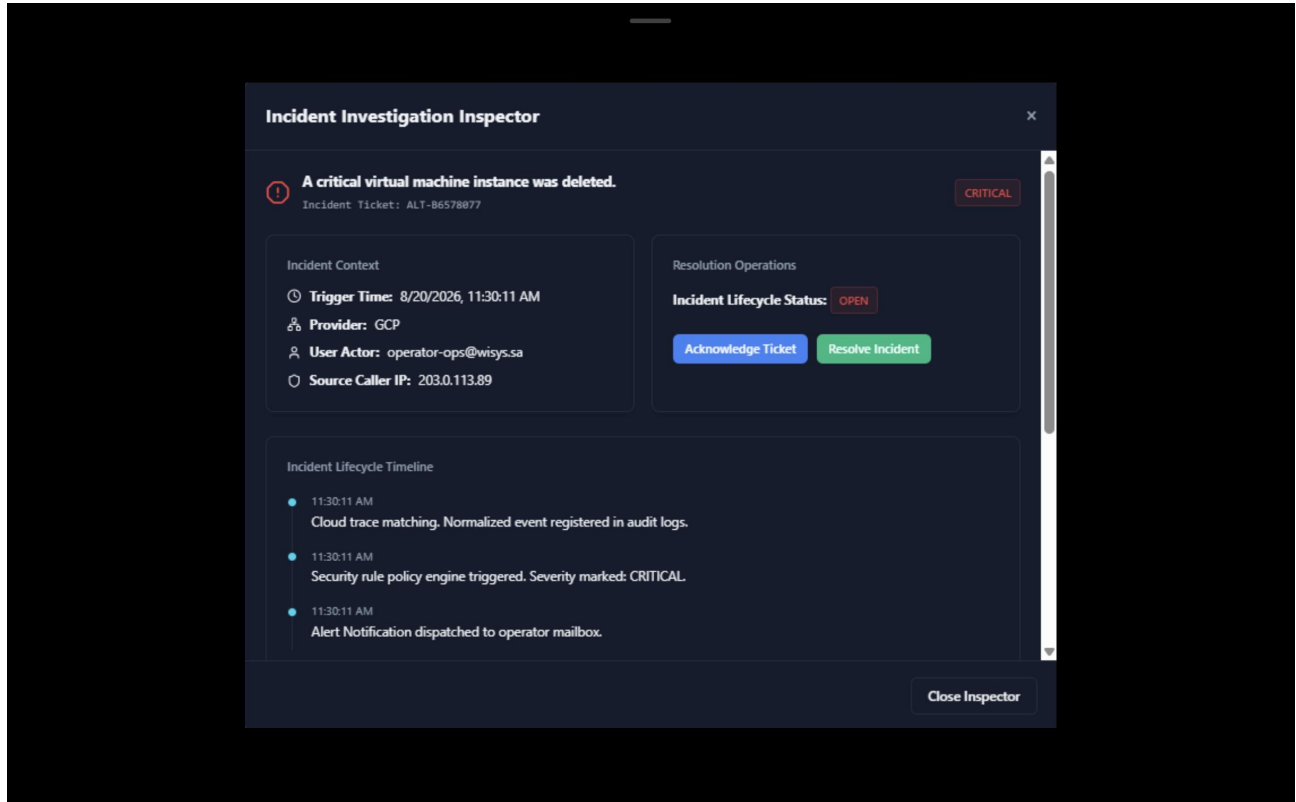
REFERENCES

- Huawei Cloud documentation – Cloud monitoring and notification services.
- Google Cloud documentation – Cloud Monitoring and Cloud Logging.
- Google Cloud documentation – IAM and audit logging.
- Google Cloud security documentation and Security Command Center concepts.
- Microsoft documentation – Microsoft identity and authentication concepts.

APPENDIX – SIMULATOR SCREENSHOTS

Only screenshots of the independently developed simulator are intended to be included in this appendix. Screenshots of WISYS production/cloud environments are intentionally excluded due to company confidentiality and security policy.

A. VM Deleted Dashboard



B. Alerts

WISYS MULTI-CLOUD

Operations & Alerting Console

Dashboard

Resources

Alerts2

Event Simulator

Rules

Audit Logs

Security Score85%

Notifications

CSE443 Internship Evaluation

Academic Local Sandbox

Console / Alerts

ACADEMIC SIMULATION ENVIRONMENT

Security Incident Alerts Console

Actionable alarms matched by the security rule engine

Search by resource, actor, description...

All Severities

All Providers

All Statuses

ALERT ID	CLOUD PROVIDER	CATEGORY	SEVERITY	RESOURCE	EVENT TYPE	DESCRIPTION	STATUS	ACTIONS
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C. Audit Logs

WISYS MULTI-CLOUD

Operations & Alerting Console

Dashboard

Resources

Alerts7

Event Simulator

Rules

Audit Logs

Security Score55%

Notifications

CSE443 Internship Evaluation

Academic Local Sandbox

Console / Audit Logs

ACADEMIC SIMULATION ENVIRONMENT

System Audit Logs

Immutable ledger of administrative configuration changes and incident actions

Search by action, resource description...

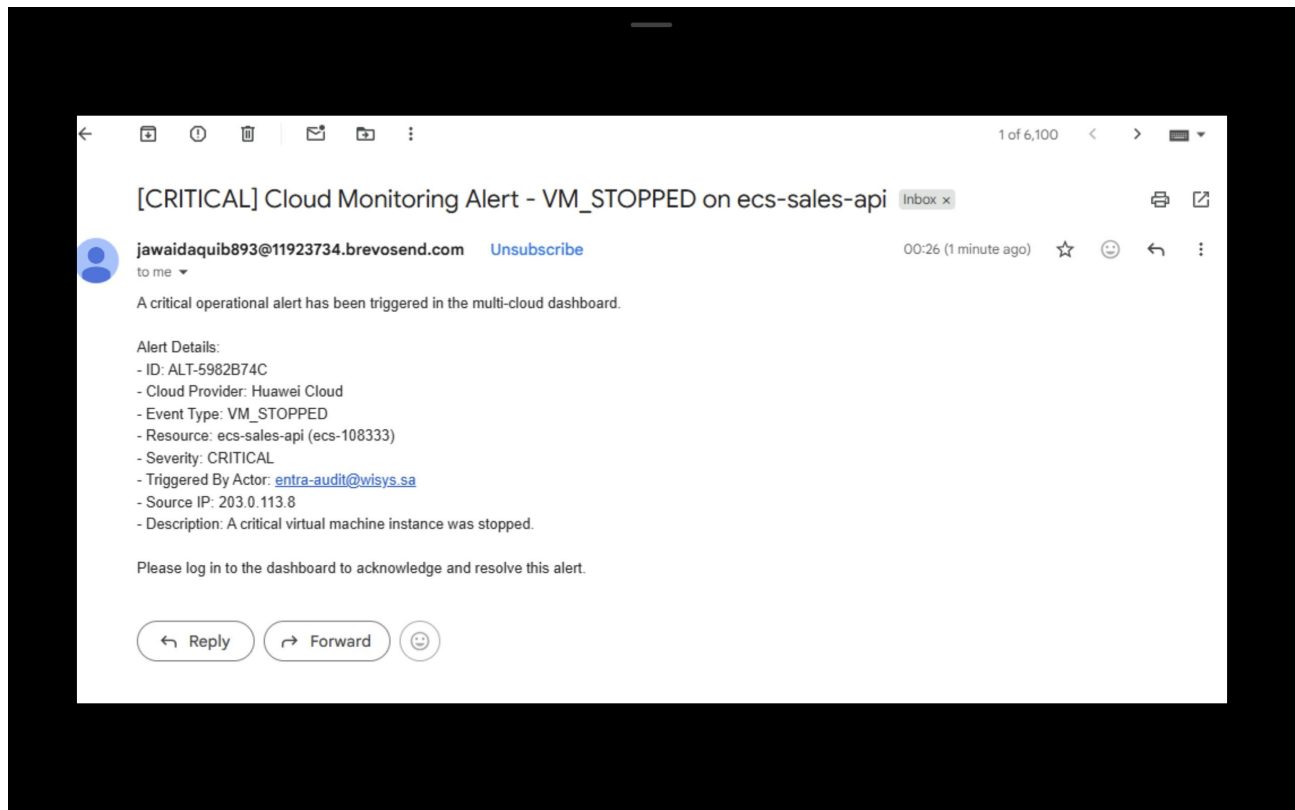
Filter by Actor Email...

All Cloud Providers

Operations Access Log

TIMESTAMP	ACTOR / INITIATOR	CALLER IP ADDRESS	ACTION OPERATION	IMPACTED RESOURCE	CLOUD TENANT	STATUS CODE
8/19/2026, 6:56:36 PM	entra-audit@wisys.sa	192.0.2.25	Huawei CTS action deleteVpc executed by entra-audit@wisys.sa on vpc-hw-prod.	vpc-hw-prod	HUAWEI CLOUD	SUCCESS
8/19/2026, 6:56:23 PM	operator-ops@wisys.sa	203.0.113.89	Huawei CTS action deleteVpc executed by operator-ops@wisys.sa on vpc-hw-prod.	vpc-hw-prod	HUAWEI CLOUD	SUCCESS
8/19/2026, 6:56:16 PM	huawei-admin@wisys.sa	203.0.113.8	Microsoft Entra ID audit log: Update application - Certificates and secrets management on application WiSys Azure Sync Integration App.	WiSys Azure Sync Integration App	MICROSOFT ENTRA ID	SUCCESS
8/18/2026, 6:33:14 PM	entra-audit@wisys.sa	203.0.113.8	GCP event VPC_DELETED detected on resource vpc-gcp-production.	vpc-gcp-production	GCP	SUCCESS
8/18/2026, 6:32:03 PM	gcp-engineer@wisys.sa	192.0.2.25	GCP event VPC_DELETED detected on resource vpc-gcp-production.	vpc-gcp-production	GCP	SUCCESS

D. Real-Time Email Notifications



E. Final Project Demonstration

